

DSS chapter 1

- **System:** an interrelated set of components including people, activities, technology, and procedures that are designed to achieve a predefined purpose.
- **Management:** is a process which organizational goals are achieved through the use of resources

NOTES

-Management is decision making

-The manager is a decision maker

-MANAGERS are always responsible for decision making

- **Decision:** is a choice made between alternative actions in a situation of uncertainty.
- **Knowledge:** awareness or familiarity gained by experience of a fact or situation.
- **Information:** defined as processed data is an increment in knowledge.
- **Data:** are only the raw facts and the material for obtaining information.
- **Database:** is an organized collection of interrelated data that reflect a major aspect of a firm's activities.
- **Downsizing:** transferring some or all the organization computing from centralized processing on mainframes or minicomputers to systems built around networked microcomputers (often in a client/server configuration).

➤ **QUALITY INFORMATION NEEDS TO HAS SEVERAL ATTRIBUTES**

- **Timely:** Available when needed and not outdated when made available
- **Complete:** Includes all the user needs to know about the situation where the information will be used
- **Concise:** Does not include elements unneeded by the user
- **Relevant:** Has direct bearing on the situation
- **Precise:** Offers quantitative information with a degree of exactness
- **Form:** The level of detail, tabular versus graphic display, and quantitative versus qualitative form are selected

Effectiveness

measures the extent to which the system meets its objectives.

Efficiency

measure of resources consumed to produce given outputs

note → Fewer resources = more efficient



-Information systems consist of (•Hardware •Software •Databases •Human resources • Procedures)

➤ **several types of information systems:**

1) Transaction Processing Systems (TPS)

- Transaction: is an activity conducted during business operations.
- TPS may work in **batch mode**, processing accumulated transactions at a single time later
on-line mode, processing incoming transactions immediately.

2) Management Reporting Systems (MRS)

- The objective of management reporting systems is to provide routine information to managers.

3) Decision Support Systems (DSS)

- Decision support systems directly support a decision-making session.
- These systems facilitate a dialog between the user, who is considering alternative problem solutions

4) Executive Information Systems (EIS)

- Executive information systems support top managers with summarized information

EIS are used to:

- Monitor the performance of the organization.
- Assess the business environment.
- Develop strategic directions for the company's future.

5) Office Information Systems (OIS)

- The main objective of OIS is to facilitate communication between the members of an organization and between the organization and its environment.

OIS are used to:

- Help to manage electronic documents
- Handle messages, such as electronic mail, facsimile, and voice mail
- Facilitate teleconferencing and electronic meetings.
- Facilitate the use of the Internet for communication and access to information.
- Facilitate the use of task-oriented teams using groupware

6) Professional Support Systems

7) Expert systems (ES)

- **Expert systems:** are system that use knowledge about its application domain and uses a reason procedure to solve problems that require human expertise

➤ **The 3 primary management levels are:**

1. Strategic planning level
2. Management control level
3. Operational control level

➤ **THE FUNCTIONS OF MANAGEMENT INCLUDE:**

- ❑ **Planning:** establishing goals and selecting the actions needed to achieve
- ❑ **Organizing:** establishing worker relationships and allows workers to work together to achieve their organizational goals.
- ❑ **Leadership:** including the people in the organization to contribute to its goals.
- ❑ **Staffing:** selecting employees for positions within the company
- ❑ **Controlling:** measuring performance against the planned objectives and take corrective action, if needed

• **DECISION MAKING:** Process of choosing amongst alternative action for the purpose of attaining a goal or goals.

: is a cognitive process that results in selection of action among several alternative scenarios

The four phases of the decision process are:

- Intelligence
- Design
- Choice
- Implementation

Note➔ DSS is distinct from the other information system and is specifically made for managers.

➤ DECISION MAKING PROCESS

1. Identification of the purpose of the decision

2. Information gathering

- you will have to gather as much as information related to the factors and stakeholders

3. Principles for judging the alternatives

4. Brainstorming and analysing different choices

You can use of Cause-and-Effect diagrams and Pareto Chart tool.

-**Cause-and-Effect diagram:** helps you to identify all possible causes of the problem

-**Pareto chart:** helps you to prioritize and identify the causes with the highest effect.

5. Evaluation of alternatives

6. Select the best alternative

7. Execute the decision

8. Evaluate the results

➤ DSS COMPONENTS

❑ **DSS Database:** It contains data from various sources

❑ **DSS Software System:** It consists of various mathematical and analytical models that are used to analyze complex data

❑ **DSS User Interface:** It is an interactive graphical interface which makes the interaction easier between the DSS and its users.

➤ DSS MODELS EXAMPLES

- ❑ **Statistical Models:** contain a wide range of statistical functions, such as mean, median, mode, deviations etc. These models are used to establish, relationships between the occurrences event and various factors related to that event.
- ❑ **Sensitivity Analysis Models:** These are used to provide answers to (what-if) situations
- ❑ **Optimization Analysis Models:** They are used to find optimum value for a target variable
- ❑ **Forecasting Models:** They use various forecasting tools and techniques
- ❑ **Backward Analysis Sensitivity Models:** Also known as goal seeking analysis, the technique followed in these models is opposite to the technique applied in sensitivity analysis models

➤ DSS BENEFITS

- 1) Improves personal efficiency.
- 2) Speed up the process of decision making.
- 3) Increases organizational control.
- 4) Encourages exploration
- 5) Speeds up problem solving
- 6) Facilitates interpersonal communication.
- 7) improve learning or training.
- 8) Generates new evidence in support of a decision.
- 9) Creates a competitive advantage over competition.
- 10) Reveals new approaches to thinking about the problem
- 11) Helps automate managerial processes.

➤ DSS ADVANTAGES AND DISADVANTAGES

Advantages (❑ Time savings. ❑ Enhance effectiveness ❑ Improve interpersonal communication
❑ Competitive advantage ❑ Cost reduction. ❑ Increase decision maker satisfaction ❑ Promote learning
❑ Increase organizational control Disadvantages)

Disadvantages (❑ Monetary cost ❑ Overemphasize decision making ❑ Assumption of relevance
❑ Transfer of power ❑ Unanticipated effects. ❑ Status reduction. ❑ Information overload.
❑ False belief in objectivity)

➤ FRAMEWORK OF DECISIONS SUPPORT SYSTEMS (DSS CLASSIFICATIONS)

❑ COMMUNICATION-DRIVEN DSS

- Help to make a meeting, or for users to collaborate.
- Uses network and communications technologies to facilitate collaboration and communication and makes decision making faster and more productive.

❑ DATA-DRIVEN DSS

- Supports decision making by analysing time-series of data and returning new information
- It is used to query a database or data warehouse

❑ DOCUMENT-DRIVEN DSS

- The purpose is to search web pages and find documents

❑ KNOWLEDGE-DRIVEN DSS

- used to provide management advice or to choose products or services.

❑ MODEL-DRIVEN DSS

- complex systems that help analyse decisions or choose between different options.

➤ DECISION SUPPORT SYSTEM TYPES

1. Traditional DSS

Components: data management + model management + dialogue management + user interface

Output: decisions based on quantitative data

2. Knowledge-based DSS (KBDSS)

Components: DSS + Knowledge management

Output: Decisions based on quantitative & qualitative data

3. Intelligent DSS based on online analytical processing (OLAP)

Components: KBDSS + OLAP

Output: provide new hidden or unexpected trends & patterns

4. Group decision support systems (GDSS)

Components: DSS + group thinking + varying time

Output: improve the DM process

5. Organizational DSS (ODSS)

Output: improving communication and coordination among people with organizations to improve organization performance and solve problems

6. Web-based decision support system

Output: extend the DSS capabilities on a global basis

• **Business intelligence (BI)**: is a technology-driven process for analysing data and delivering actionable information that helps managers and workers make business decisions.

➤ DECISION TYPES

1. **Structured**: Repetitive and routine, and decision makers can follow a definite procedure for handling them to be efficient.
2. **Semi-Structured**: Only part of the problem has a clear answer provided by an accepted procedure
3. **Unstructured**: Each of these decisions is novel, important, and nonroutine, and there is no well-understood or agreed-on procedure for making them.

FOUR STEPS IN DECISION MAKING

1. **Intelligence**: consists of discovering, identifying, and understanding the problems occurring in the organization—why is there a problem, where, and what effects is it having on the firm
2. **Design**: involves identifying and exploring various solutions to the problem.
3. **Choice**: consists of choosing among solution alternatives
4. **Implementation**: involves making the chosen alternative work and continuing to monitor how well the solution is working

Note → the output of **Intelligence phase** is problem statement
the output of **design phase** is alternative

➤ **DECISION-MAKING UNDER CERTAINTY**

- It is assumed that complete knowledge is available so that the decision maker knows exactly what the outcome of each course action will be
- The decision-maker is viewed as a perfect predictor of the future
- Decision maker knows with certainty the consequences of every alternative or decision choice.

➤ **DECISION-MAKING UNDER UNCERTAINTY**

- decision-maker does not know, or cannot estimate, the probability of occurrence of the possible outcomes.
- Decision-making under uncertainty is more difficult because of insufficient information.
- decision maker does not know the probabilities of the various outcomes.
- • **Maximax (optimistic approach)** → Choose decision with the maximum of the maximum payoffs.
 - If the payoff table was in terms of costs, the decision with the lowest cost would be chosen.
 - If the payoff table was in terms of profits, the decision with the highest profit would be chosen.
- • **Maximin (conservative approach)** → Choose decision with the maximum of the minimum payoffs.
 - If the payoff was in terms of costs, the maximum costs would be determined for each decision and then the decision corresponding to the minimum of these maximum costs is selected.
 - If the payoff was in terms of profits, the minimum profits would be determined for each decision and then the decision corresponding to the maximum of these minimum profits is selected.
- • **Minimax regret (regret approach)** → Choose decision with the minimum of the maximum regrets for each alternative

➤ **DECISION-MAKING UNDER RISK**

- The decision-maker must consider several possible outcomes for each alternative, each with a given probability of occurrence
- The decision maker knows the probabilities of the various outcomes

➤ **decision problem is characterized by decision alternatives, states of nature and resulting payoffs.**

❑ **decision alternatives:** are the different possible strategies the decision maker can employ.

❑ **states of nature:** refer to future events, not under the control of the decision maker, which will affect decision results.

➤ **Decision theory problems are represented as one of the following:**

- **Payoff Table**

- **payoff:** The result from combination of a decision alternative and a state of nature

- **Decision Tree**

DSS chapter 4

- **Model-driven decision support system (MDDSS):** utilizes a model to solve problems or aid in decision making.

Note→ A model driven DSS may employ a single model or a combination of two or more models

Note→ Model-driven DSS are generally not data intensive. Rather they use parameters entered by decision makers and help them analyse a situation.

- **Modelling:** is the process of identifying an appropriate model for a prospective model- driven decision support system

Note→ Model validation is done by comparing model's output and actual behaviour of the event.

➤ **Model-driven DSS are designed assuming any of the analyses – static and dynamic.**

❑ **Static Analysis:** doesn't take into consideration the long-term response of a system.

: It takes a single snapshot of a situation and assumes that it will remain stable all through and won't change.

❑ **Dynamic Analysis:** is testing a program or a software system in real-time.

- This method considers that the situation changes over time, due to any reason

➤ **A DSS deploys one or combination of below models:**

❑ **Explanatory/Descriptive Model:** Describes and explains why something is the way it is and why and how it works.

❑ **Contemplative Model:** Forecasts results or outcomes that may be produced from a specific set of parameters.

❑ **Algebraic Model:** A high-level modelling system for solving complex equations

➤ **ACCOUNTING AND FINANCIAL MODELS**

1- BREAK-EVEN ANALYSIS: is a financial tool that helps businesses determine the sales volume or revenue

❑ A break-even analysis model typically includes the following components:

- I. **Fixed costs:** These are costs that remain the same regardless the level of production
- II. **Variable costs:** These are costs that change with the level of production
- III. **Selling price:** The price at which the product or service is sold.

❑ Break-even point = Fixed costs / (Selling price - Variable cost per unit)

2- BUDGET FINANCIAL MODEL

❑ budgeting model can help managers to:

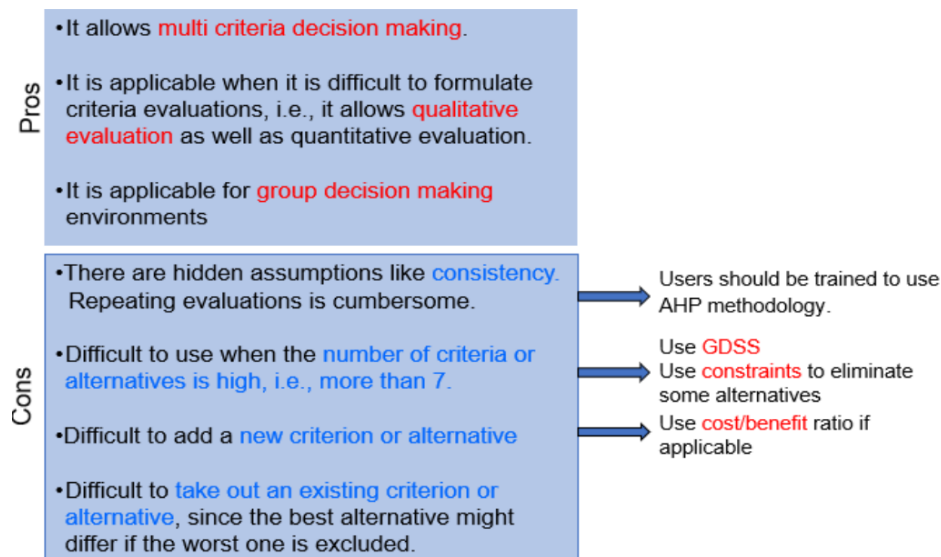
- Set financial goals and objectives.
- Develop and allocate resources.
- Track and analyse financial performance.
- Forecast future financial performance.

3- PRO FORMA FINANCIAL STATEMENT: can summarize the anticipated financial results for a specific period in the future.

4- RATIO ANALYSIS

➤ **DECISION ANALYSIS MODELS:** identify and evaluate alternatives with their respective pros and cons.

1- Analytical Hierarchy Process (AHP): It's a multi-criteria decision technique (MCDM) that combines quantitative and qualitative factors when evaluating alternatives.



2- DECISION TREES: aim is to identify the most appropriate strategy to reach a goal.

- **A decision tree has three types of nodes.**

- **Choice node:** represented by a square

- **Chance node:** represented by a circle.

- **End node:** represented by a triangle.

3- MULTI-ATTRIBUTE UTILITY ANALYSIS (MAUA)

Note→ MAUA emphasizes the quantification of individual preferences using utility functions, while AHP relies on pairwise comparisons and the determination of relative importance among criteria and alternatives.

4- INFLUENCE DIAGRAMS: It's a diagrammatic representation of a decision situation, to express relationships between variables and the influence of variable on others.

❖ **The elements of an influence diagram are:**

- **Decision nodes:** represented by squares and represent the decisions that need to be made.

- **Chance nodes:** represented by circles and represent the uncertainties in the problem.

- **Value nodes:** represented by hexagons and represent the objectives of the decision-maker.

- **Arcs:** represent the relationships between the different elements of the problem.

➤ **FORECASTING MODELS:** is to predict the value of interrelated variables at some point of time in future.

❖ **The two main types of forecasts are:**

✓ **Short run forecasts:** where the prediction will be used anytime soon

✓ **Long run forecasts:** where the prediction is used for long term investment/planning decisions.

❖ **VARIOUS TYPES OF FORECASTING MODELS**

1- Naïve Exploration: simple forecasting that provides limited accuracy.

2- Judgment Methods: based on the perceptions and opinions of experts instead

- 3- Moving Average:** the predictions are based upon the historical values
 - 4- Exponential Smoothing:** it alters the historical data mathematically to reflect the assumptions of a decision maker
 - 5- Time Series Extrapolation:** considers the economic variables that are measured at consecutive intervals of time
 - 6- Regression and Econometric Models:** use of linear and multiple regressions to establish cause and effect relationships
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➤ **NETWORK AND OPTIMIZATION MODELS:** are integrated into a DSS when decisions regarding resource allocation, project control, location, scheduling,

➤ **SIMULATION MODELS:** conduct experiments to identify conditions or situations that near from the actual conditions.

➤ **MATHEMATICAL MODELS :** made up of three basic components: decision variables, uncontrollable variables and result variables

➤ **Knowledge-Driven DSS:** is a computer-based reasoning system that provides information, comprehension and suggestions to users to support them in decision-making.

•**Expert System:** A computer system that imitates the decision-making capability of a human expert is called an expert system or an artificial intelligence system.

It is designed to solve problems by

- Using if-then rules
- Reasoning about knowledge
- Drawing inferences from facts and rules

•**Knowledge Discovery and Data Mining:** These are interrelated terms used for the process of extracting valuable knowledge and discovering patterns

❖ two types of data mining models:

Predictive Model: This model predicts which prospects are likeliest to respond to a particular stimulus. It forecasts clear results based on patterns identified from known results.

Descriptive Model: This model describes patterns in existing data to create significant demographics subgroups

➤ WEB-BASED AND INTER-ORGANIZATIONAL DSS

Note→ web-based decision support systems can be knowledge- driven, communications-driven, model-driven, document-driven or data-driven or hybrid

❖ EXAMPLES OF WEB-BASED DSS

- **Microsoft CarPoint:** a web-based system that allows users to make comparisons between car models.
 - **TCB Works:** A communication-driven web-based DSS, TCB Works enables people to interact, discuss and make structured decisions.
 - **Fidelity 'Retirement Planning Calculator':** It is a model-driven DSS to help a person decide how much needs to invest for retirement.
 - **Netscape Decision Guides:** These are model and knowledge-driven webbased decision support systems, offering 25 different decision guides on diverse topics
 - **Stockfinder:** It's a data-driven DSS to help investors identify stocks
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DSS chapter 5

- **database:** is a collection of related data
- **data:** is a collection of facts and figures that can be processed to produce information.
- **database management system:** stores data in easy way to retrieve, manipulate, and produce information

• **data dictionary:** (also called the metadata) is the data about the data.

- ✓ **active data dictionary:** (also called integrated data dictionary) is managed automatically by the database management software. Consistent with the current structure and definition of the database.
- ✓ **passive data dictionary:** (also called non-integrated data dictionary) is used only for documentation purposes. It is managed by the users of the system and is modified whenever the structure of the database is changed.

Note → Data dictionary provides the name of a data element, its description, and data structure

• **Interface:** the part of a computerized system that allows its users to interact with it.

: It is what users see and use.

• **ROMC:** is an organized approach to design specialized decision support systems, more precisely their user interfaces.

• **Representation:** is presenting information or results in a structured way.

• **Control aid:** is provided to users, so that they can effectively use representations, operations and memory aids.

Note → A data warehouse is its memory aid and so for decision makers. So, it

DSS chapter 6

➤ Designing and development of a decision support system is a complex process; you may face several issues:

- 1) Mismatch between the perspectives of the programmer and decision maker
- 2) Identification of requirements
- 3) Technology selection
- 4) Approach to software design and development

➤ Most of these issues can be resolved by:

- 1) Identifying clear agendas
- 2) Brainstorming with team members
- 3) Communicating the expectations clearly to the programmers
- 4) Educating about various technologies used in DSS development.
- 5) Trying to reach a common platform with the developers.

➤ DSS DESIGN PROCESS

1- INITIALIZATION

- ☐ involves defining the purpose and objectives of the DSS.
- ☐ identify the decision-making problems that the DSS will address.

2- DECISION-ORIENTED DIAGNOSIS

- ☐ conduct a thorough analysis of the current decision-making processes within the organization.
- ☐ This step often involves interviews, surveys, and data analysis
- ☐ **Diagnosis:** is the identification of problems or opportunities for improvement in decision behaviour.

Note → Diagnosis involves **determining** how decisions are currently made, **specifying** how decisions should be made, and **understanding** why decisions are not made as they should be.

☐ **audit:** refers to a systematic and comprehensive examination or review of the Decision Support System

Note → primary purpose of an audit is to identify strengths and weaknesses in the DSS

☐ **Decision Support System (DSS) audit plan:** is a structured approach for evaluating the effectiveness, efficiency, and security of a Decision Support System

3- PREPARE A FEASIBILITY STUDY

- ☐ should occur prior committing resources to developing DSS.
- ☐ Involves assessing the technical, economic, operational, and scheduling feasibility of the project

4- MAKE OR BUY

- ☐ In this step, you decide whether to build the DSS in-house or buy existing DSS solution from a vendor.

Note→ Buying a DSS may save time and resources, while building one in-house allows for more customization and control.

Note→ Packaged DSS applications are less expensive to implement than in-house development. and also often faster to implement.

❖ three main approaches to system design and development:

1) system development life cycle (SDLC) approach

is based on a series of formal steps, including the following seven steps:

- 1) Confirm user requirements.
- 2) Systems analysis.
- 3) System design.
- 4) Programming.
- 5) Testing.
- 6) Implementation.
- 7) Use and Evaluation

2) Rapid prototyping

- ☐ Creating a rapid prototype for (DSS) system involves building a simplified version of the system to demonstrate its features and functionality.
- ☐ Create rapid prototype for a DSS allows you to visualize and test the system's design early
- ☐ **prototyping methodology usually includes five steps:**
 1. Identify user requirements.
 2. Develop a first iteration DSS prototype.
 3. Evolve and modify the next iteration DSS prototype.
 4. Test DSS and return to step 3 if needed.
 5. Full implementation.

3) End-user approach.

- ☐ emphasizes active involvement of end-users in the system development process.
- ☐ End-user DSS development is a controversial

➤ ESS

- **ESS:** help managers with unstructured and semi structured problems by focusing on the needs of senior management.

- ☐ Designed for top-level executives and senior managers.

- ☐ primary purpose is to provide high-level, summarized information to help in strategic decision-making.

➤ ESS COMPONENTS

- ☐ **The knowledge base:** This is where the information is stored in the expert system i

- ☐ **The user interface:** This is where the user interacts with the expert system.

- ☐ **The inference engine:** This applies the facts to the rules and determines the questions to be asked of the user

- **EXECUTIVE SUPPORT SYSTEMS(ESS):** Bring data from all parts of the firm and enable managers to select, access, and tailor them as needed.

➤ ESS can help managers to :

- 1- Identify changing market conditions,
- 2- Formulate responses
- 3- Track implementation efforts
- 4- Learn from feedback.

NOTE→ ESS Include tools for modelling and analysis.

DSS CHAPTER 8

❑ A system that integrates knowledge from experts is commonly called **knowledge-based decision support system (KBDSS)** or **intelligent decision support system (IDSS)**

• **AUTOMATED DECISION SYSTEMS (ADS)** : is a rule-based system that provides a solution usually in one functional area to a specific repetitive managerial problem

NOTE→ ADS suitable for frontline employees

➤ ARTIFICIAL INTELLIGENCE

• **Artificial intelligence (AI)**: is a collection of concepts and ideas that are related to the development of intelligent systems.

: is the study of how to make computers do things at which, at the moment, people are better.”

❖ **AI is concerned with two basic ideas:**

- 1) The study of human thought
- 2) The representation and duplication of those thought processes in machines

➤ EXPERT SYSTEMS

• **Expert systems (ES)**: are computer-based information systems that use expert knowledge to achieve high-level decision

Note→ MYCIN is the most well-known ES application

• **Expert**: is a person who has the special knowledge, judgment, experience, and skills to put it in action to provide advice and to solve complex problems

❑ **Decision performance** and **level of knowledge a person has** are criteria used to determine a person is expert or not

• **Expertise**: is the extensive, task-specific knowledge that experts has.

❖ **FEATURES OF Expert systems (ES):**

- Expertise
- Symbolic reasoning.
- Deep knowledge.
- Self-knowledge

❖ **ES can be viewed as having two environments:**

ES builder uses

• **development environment:** to build the necessary components of the ES and to populate the knowledge base with expert knowledge.

nonexpert uses

• **consultation environment:** to obtain advice and to solve problems

➤ **Main Components of an Expert System (ES):**

1. Knowledge Base

- is The foundation of any expert system.

knowledge base include two elements:

Facts: Describe the specific problem situation.

Heuristics/Rules: Represent expert knowledge for solving domain-specific problems.

2. Inference Engine

- The “brain” of the expert system.
- Also called the control structure or rule interpreter.
- Uses knowledge and blackboard data to reason and draw conclusions.

3. User Interface

• **Knowledge Acquisition Subsystem**

- Involves gathering, transferring, and transforming problem-solving expertise from experts or reliable sources to the system.

• **Blackboard (Workplace)**

- area of working memory for the current problem description.
- used for recording intermediate results, hypotheses, and decisions.

Blackboard Records three types of decisions:

- Plan (how to attack the problem)
- Agenda (pending actions)
- Solution (current hypotheses or alternative actions)

• **Explanation Subsystem (Justifier)**

- The ability to trace responsibility for conclusions.

Answers questions like:

Why was a question asked?

How was a conclusion reached?

Why was an option rejected?

What is the full decision path for reaching the conclusion?

- **Knowledge-Refining System**

- Analyses successes and failures. And Evaluates system performance to improve accuracy and reasoning.
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- **Knowledge Engineering**

- Collection of activities encompassing the acquisition knowledge from human experts and conversion this knowledge into a repository

Two perspectives:

Narrow: Focused on building expert systems (acquisition, representation, validation, inferencing, explanation).

Broad: describes the entire process of developing and maintaining intelligent systems

Note → **production rules** is most popular form of **knowledge representation** for expert systems.

➤ **Knowledge from experts needs to be evaluated for quality, including evaluation, validation, and verification**

❑ **Evaluationis:** objective is to assess an ES's overall value. .

❑ **Validationis:** the part of evaluation that deals with the performance of the system

❑ **Verificationis:** building the system right or substantiating that the system is implemented to its specifications.

➤ **Two types of rules are common in artificial intelligence: knowledge and inference.**

❑ **Knowledge:** also called declarative rules, state all the facts and relationships about a problem.

❑ **Inference:** also called procedural rules, offer advice about how to solve a problem

Note → Inference rules contain rules about rules and thus are also called meta-rules.

❖ **The knowledge engineer separates the two types of rules:**

-**Knowledge rules** go to the knowledge base,

-**inference rules** become part of the inference engine

•**Inferencing (or reasoning):** is the process of using the rules in the knowledge base with the known facts to draw conclusions.

➤ **The most popular inferencing mechanisms for rule-based systems are forward and backward chaining:**

➤ **Backward chainingis:** a goal-driven approach, we start from an expectation of what is going to happen

➤ **Forward chainingis:** a data-driven approach. We start from available information as it becomes available or from a basic idea

Note → Having knowledge implies that it can be exercised to solve a problem, whereas having information does not carry the same connotation

• **Knowledge management:** is a process that helps organizations identify, select, organize, and transfer important information

• **Knowledge management (Km):** is the systematic and active management of ideas, information, and knowledge for organization's employees.

➤ KM approaches

1- Process approach

- **process approach:** knowledge management attempts to codify organizational knowledge
- Organizations that adopt the process approach may implement explicit policies
- This approach is favored by firms that sell standardized product
- In these cases, the knowledge may be typically static in nature.

2- practice approach

- **practice approach:** assumes that a great deal of organizational knowledge is tacit
- adopted by companies that provide highly customized solutions to unique problems.
- In these cases, the knowledge is tacit in nature

• **Groupwork:** refers to work done by two or more people together.

• **GROUP DECISION SUPPORT SYSTEMS (GDSS):** An interactive computer-based system that facilitates the solution of semi structured or unstructured problems by a group of decision makers.

- **The goal of GDSS** is to improve the productivity of decision-making meetings

• **A group support system (GSS):** is any combination of hardware and software that enhances groupwork

➤ techniques are available for designing intelligent systems that can learn by themselves
These include :

- ❑ **case-based system:** contains a case base that maintains expert experience.
- ❑ **Artificial neural networks:** use artificial neurons that executable on computers to mimic the human brain.
- ❑ **Genetic algorithms:** modify the natural-selection process to find useful knowledge.
- ❑ **Fuzzy systems:** convert between symbolic reasoning and mathematical calculation to improve decision performance

• **machine learning:** learn from past experience to improve their own performance

: is a family of methods that attempt to allow machines to gain knowledge for problem-solving by showing them historical cases.

Note → Machine learning is nontrivial.

➤ MACHINE LEARNING METHODS

❑ **Supervised machine learning:** also called target variable. is used for making predictions from data. To be able to do that, we need to know what to predict

❑ **Reinforcement Learning (RL):** is the science of decision making. It is about learning the optimal behaviour in an environment to obtain maximum reward.

❑ **Unsupervised learning:** is self-organized learning. Its main aim is to explore the underlying patterns and predicts the output

MACHINE LEARNING METHODS

Criteria	Supervised ML	Unsupervised ML	Reinforcement ML
Definition	Learns by using labelled data	Trained using unlabelled data without any guidance.	Works on interacting with the environment
Type of data	Labelled data	Unlabelled data	No – predefined data
Type of problems	Regression and classification	Association and Clustering	Exploitation or Exploration
Supervision	Extra supervision	No supervision	No supervision
Algorithms	Linear Regression, Logistic Regression, SVM, KNN etc.	K – Means, C – Means, Apriori	Q – Learning, SARSA
Aim	Calculate outcomes	Discover underlying patterns	Learn a series of action
Application	Risk Evaluation, Forecast Sales	Recommendation System, Anomaly Detection	Self Driving Cars, Gaming, Healthcare